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M1 MACRO

Group B

Session 17

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Habit persistence GE model

Firms $\rightarrow$ Representative firm [competitive]

$$\max_{\{y_t, h_t\}} \pi_t = P_t y_t - w_t h_t$$

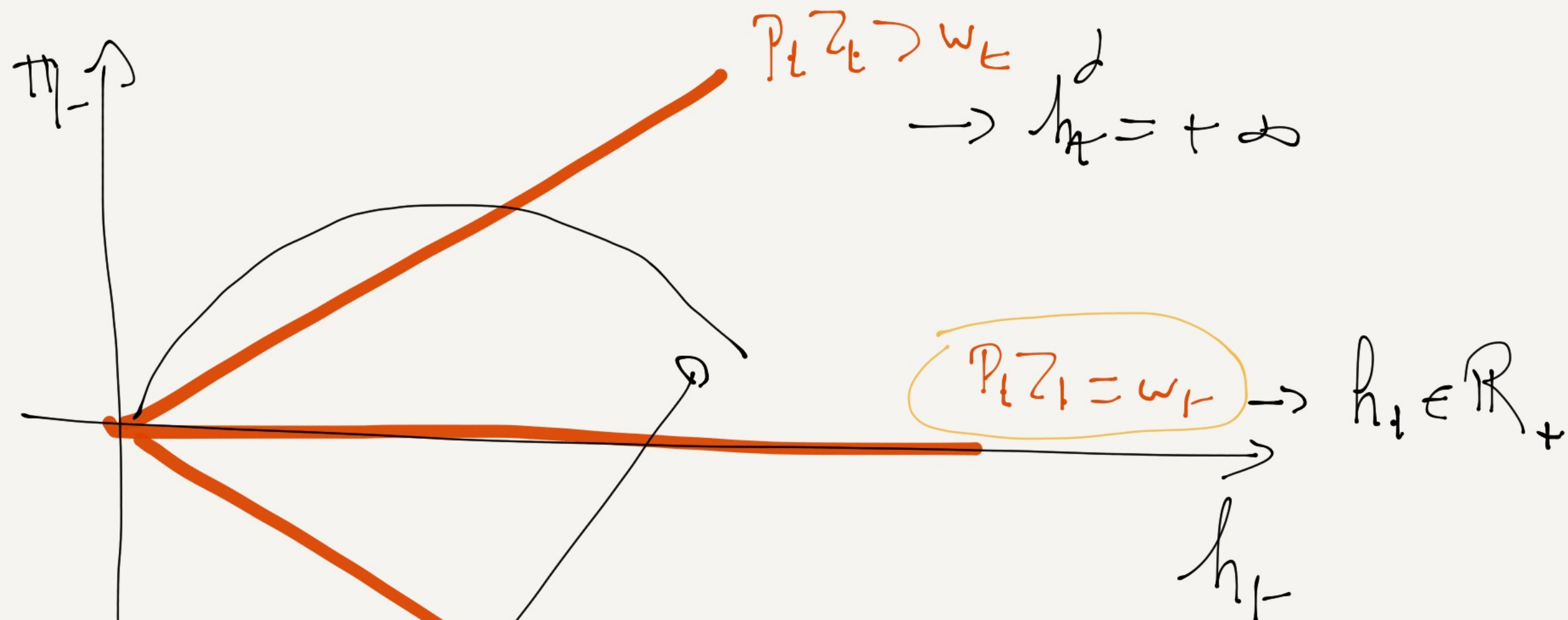
$$\text{s.t. } y_t = z_t h_t$$

productivity
(exog)

$$\Rightarrow \max_{\{h_t\}} P_t z_t h_t - w_t h_t = (P_t z_t - w_t) h_t$$

FOC : $\boxed{P_t z_t = w_t}$

$\leftarrow$ badly written



$$[if \eta_t = Z_t h_t^d, 0 < \alpha < 1]$$

$$h_t^d = \begin{cases} +\infty & \text{if } P_t Z_t > w_t \\ \in \mathbb{R}^+ & \text{if } P_t Z_t = w_t \\ 0 & \text{if } P_t Z_t < w_t \end{cases}$$

and $y_t = Z_t h_t^d$

Hh Representative (competitive)

$$M = E_0 \sum \beta^t \left[\underbrace{\log(c_t - b c_{t-1})}_m - \underbrace{h_t}_{\text{disutility of labor}} \right]$$

$$0 \leq b < 1$$

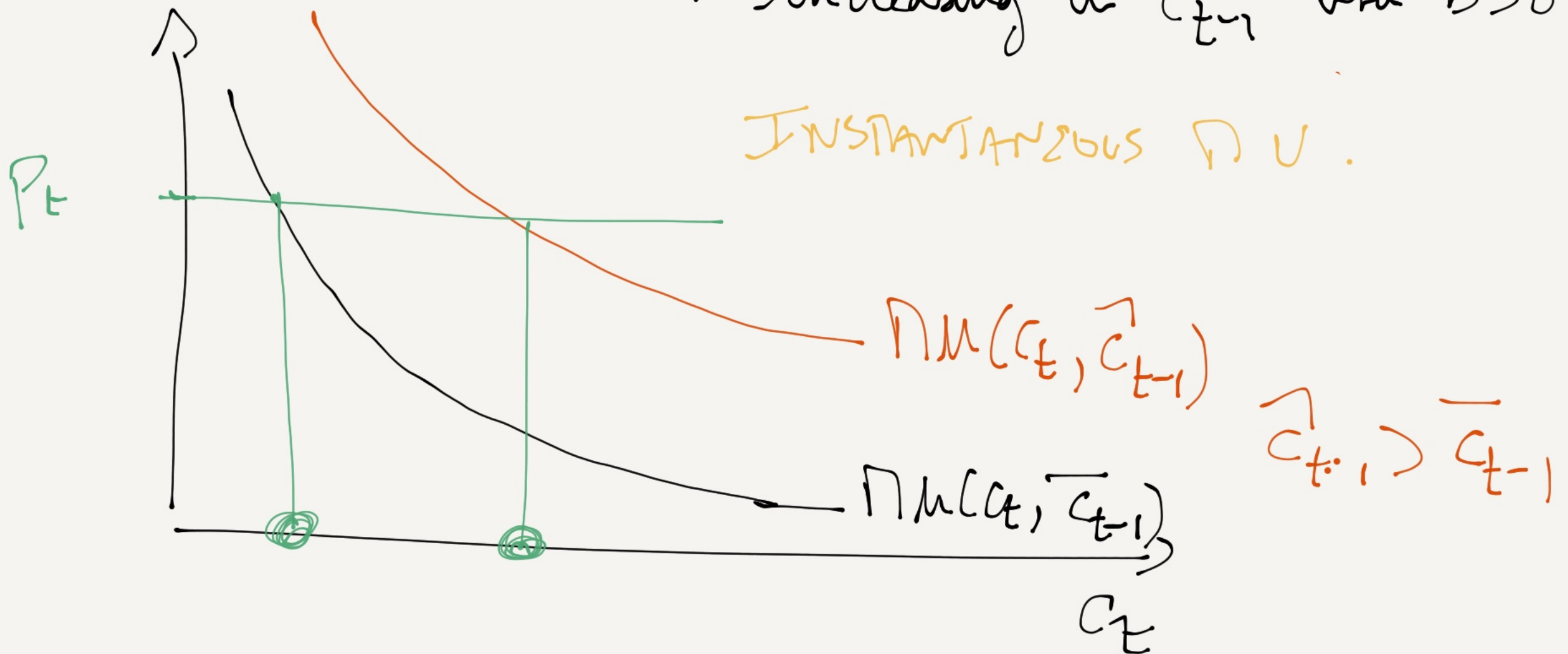
= habit persistence

[if $b=0$, back to basic model]

$$\text{MUR of } C_t : \frac{d\mu}{dC_t} = \frac{\beta^t}{C_t - bC_{t-1}}$$

↳ decreasing in C_t
 ↳ increasing in C_{t-1} when $b > 0$

INSTANTANEOUS $\cap U$.



Budget constraint $P_t C_t = w_t h_t + \pi_t, \forall t$
 [No savings]

$$C_t = \frac{w_t h_t + \pi_t}{P_t} \rightarrow \text{plug in } M.$$

$$M = E_0 \sum \beta^t \left(\log \left(\frac{w_t h_t + \pi_t}{P_t} - b \frac{(w_{t-1} h_{t-1} + \pi_{t-1})}{P_{t-1}} \right) - h_t \right)$$

$$\max_{\{h_t\}} M$$

FOC wrt h_t

FOC wrt h_t in period t

$$p_t \left(\frac{w_t}{p_t} \times \frac{1}{c_t - b c_{t-1}} - 1 \right) - \beta \frac{p_{t+1} b \frac{w_{t+1}}{p_{t+1}}}{c_{t+1} - b c_t} = 0$$

$$\Rightarrow \boxed{\frac{1}{c_t - b c_{t-1}} - \beta \frac{p_{t+1} b}{c_{t+1} - b c_t} = \frac{1}{w_t / p_t}}$$

c_0 given
+ transversality condition

G.E : A competitive GE is

(c_t, y_t, h_t^s, h_t^d) and (P_t, w_t) s.t

• for given prices, firms and Hh max. objectives

• markets clear

$$\begin{cases} c_t = y_t \\ h_t^s = h_t^d \end{cases}$$

Note

• Walras' law : $P_t = 1$ [numeraire]

• labor market eq $\Rightarrow P_t Z_t = w_t$

$$\Rightarrow w_t = Z_t$$

Euler eq. at equilibrium

$$\frac{1}{c_t - b c_{t-1}} - \beta b E_t \frac{1}{c_{t+1} - b c_t} = \frac{1}{z_t} \quad (a)$$

change of variable : $X_t = \frac{1}{c_t - b c_{t-1}}$

(a) $\Rightarrow X_t - \beta b E_t X_{t+1} = \frac{1}{z_t}$

$$0 < \beta b < 1$$

Solve forward

$$X_t = \frac{1}{z_t} + \beta b \bar{E}_t X_{t+1}$$

$$= \frac{1}{z_t} + \beta b \bar{E}_t \left[\frac{1}{z_{t+1}} + \beta b \bar{E}_{t+1} X_{t+2} \right]$$

$$= \frac{1}{z_t} + \beta b \bar{E}_t \left(\frac{1}{z_{t+1}} \right) + (\beta b)^2 \bar{E}_t X_{t+2}$$

$$X_t = \sum_{j=0}^{\infty} (\beta b)^j \bar{E}_t \left(\frac{1}{z_{t+j}} \right) + \lim_{j \rightarrow \infty} (\beta b)^j \bar{E}_t X_{t+j}$$

$= 0$ (Transversality)